Chipsee

Rugged Laptop

PPC-156-B1X



15.6” Fully Rugged Laptop

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PPC-156-B1X



Front View



Rear View



Side View 1



Side View 2

1 Product Overview

The PPC-156-B1X fully rugged laptop is a 15.6” laptop built for the most demanding environments. It is powered by a 12th to 14th Gen Intel® Core™ processor, optionally with an NVIDIA® graphics card, and its sunlight-readable display stays readable outdoors.

The rugged aluminum alloy casing is IP65 rated against dust and water, and the laptop works reliably in extreme temperatures. Optionally, it can be made resistant to mold, salt spray and damp heat, and to transportation vibration, vibration and shock, and it can comply with MIL-STD EMC requirements. A wide range of communication modules can be built in.

2 Ordering Options

Most of the Chipsee products can be customized during the ordering process. The product will be shipped with the factory defaults if no extra requirements are specified. The table in the Hardware Features section lists the default and optional features of the product.

> **Note**
>
> You can order PPC-156-B1X from the official Chipsee Store or from your nearest distributor.

2.1 Operating System

PPC-156-B1X supports Microsoft® Windows 10/11 Pro 64-bit.

2.2 Optional Features

By default, PPC-156-B1X has no wireless communication module installed. Wi-Fi 6E, Bluetooth 5.2, 5G/4G LTE and GPS/BeiDou modules can be selected during the ordering process, as well as the graphics card, the touch screen, the camera and the fingerprint reader.

> **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only.
> Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

3 Hardware Features

The key features of PPC-156-B1X are listed in the table below.

Key Features

	PPC-156-B1X
CPU	12th to 14th Gen Intel® Core™ i3, i5, i7 or i9
GPU	Intel® UHD Graphics, NVIDIA® RTX 3050 8GB or RTX 3060 12GB GDDR6 **(optional)**
RAM	16GB DDR5 SO-DIMM, 2 slots, up to 64GB
Storage	1 x M.2 2280 SSD (NVMe/SATA), 512GB / 1TB / 2TB / 4TB / 8TB, and 1 x mSATA SSD, 512GB / 1TB / 2TB
Display Size	15.6”
Display Type	Sunlight-readable LCD
Display Resolution	1920 x 1080 (16:9)
Brightness	Up to 1000 cd/m^2
Touch Technology	Capacitive multi-touch **(optional)**
Glass	Tempered glass
USB	4 x USB 3.0 Type-A, 2 x USB 2.0 Type-A
LAN	2 x RJ45, 10/100/1000Mbps auto-sensing
HDMI	1 x HDMI Out
Serial Port	3 x DB9, RS232/RS422/RS485, up to 115200bps
Audio	Speakers, microphone, 1 x 3.5mm combo audio jack
WiFi/BT	Wi-Fi 6E, Bluetooth 5.2 **(optional)**

PPC-156-B1X	
4G/5G	5G/4G LTE **(optional)**
GPS	GPS, BeiDou **(optional)**
Camera	5MP HD camera **(optional)**
Fingerprint Module	Fingerprint reader **(optional)**
Keyboard	LED-backlit waterproof membrane keyboard with numeric keypad, touchpad
LED Indicator	NumLock, CapsLock, battery, power
Battery	21.6V, 10500mAh, 226.8Wh Li-ion battery
Power Input	350W AC/DC power adapter, 28V DC output (450W optional)
OS	Windows 10/11 Pro 64-bit
Dimensions	From 390.0 x 270.0 x 58.5mm
Weight	From 6.5kg, with Li-ion battery
Operating Temp.	From -40°C to +55°C (system unit, excluding battery)
Storage Temp.	From -50°C to +60°C
Shell Material	Aluminum alloy
Waterproof	IP65 (IEC 60529 compliant)
Humidity	95% RH, non-condensing
Vibration	5 to 500Hz, 1.5Grms
Shock	20G peak acceleration (11ms duration)
Environmental Resistance	Mold, salt spray and damp heat **(optional)**
Mechanical Protection	Transportation vibration, vibration and shock **(optional)**
EMC	CE101, CE102, CS101, RE101, RE102 **(optional)**

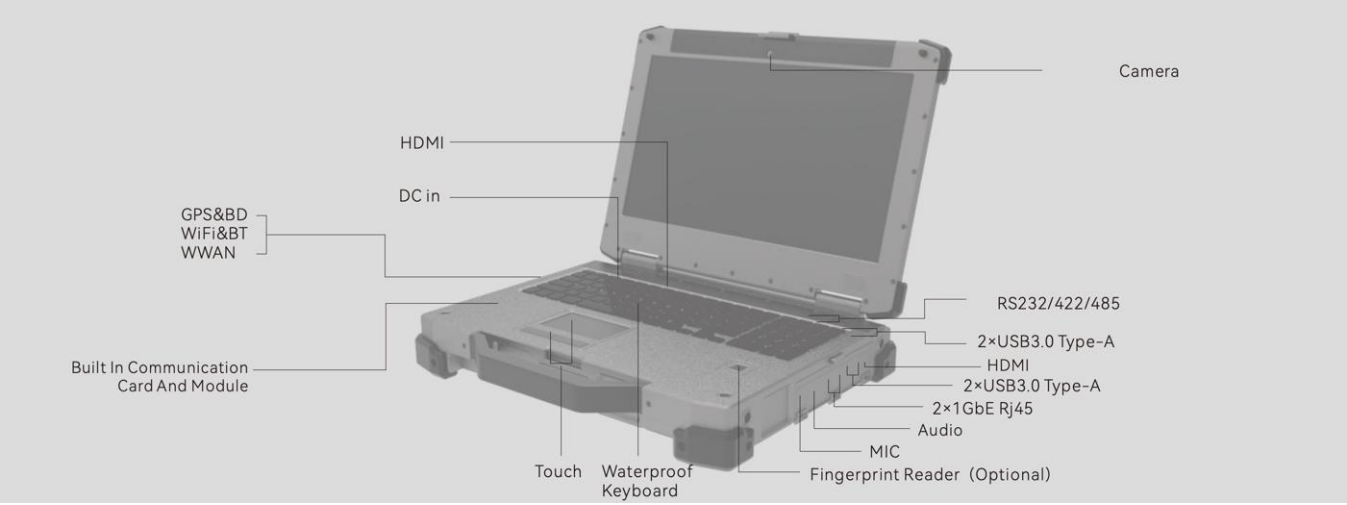
Note

The battery only operates above -20°C; its performance will degrade below 0°C.

The internal battery is rated for operation from 0°C to +55°C. For environments below 0°C, use of an external power supply or an optional low-temperature battery is recommended.

4 Interfaces

The figure below shows the interfaces of PPC-156-B1X.



Interfaces of the product

5 Accessories

Accessory	Specification	Supply
Power adapter	350W AC/DC, 28V DC output	Default
Li-ion battery	21.6V, 10500mAh, 226.8Wh	Default
Power adapter	450W AC/DC, 28V DC output	Optional
DVD drive	USB 8X DVD-RW	Optional
Shoulder strap	Quick-release X shoulder strap	Optional
Crossbody bag	Waterproof shoulder bag	Optional
Air suitcase	Sturdy PC carrying case	Optional

6 Disclaimer

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